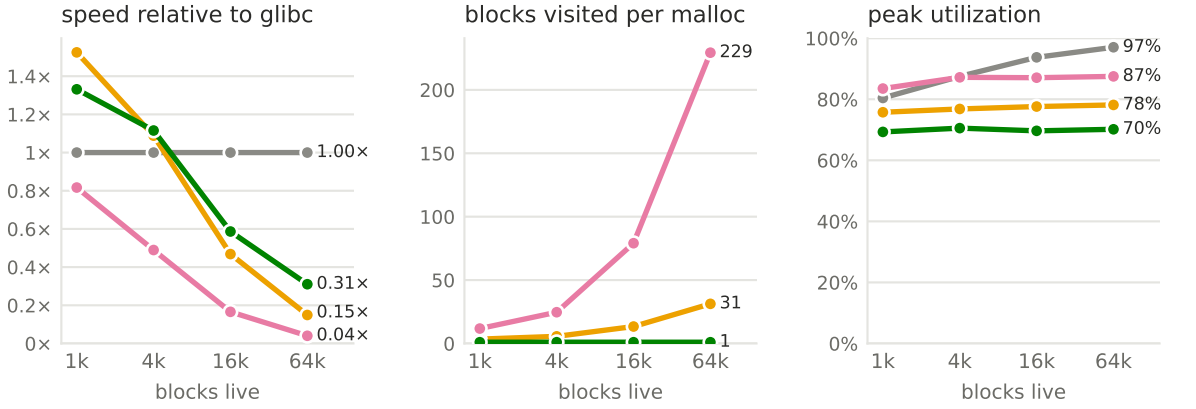The random workload with a growing live set